

Series of real numbers

1) Calculate the sum of the following series:

$$a) \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$$

$$b) \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$$

$$c) \sum_{n=0}^{\infty} \frac{1}{n(n+1)\dots(n+p)} \quad | p \geq 1$$

$$a) \quad a_n = \frac{1}{4n^2 - 1} = \frac{1}{(2n-1)(2n+1)} = \frac{1}{2} \cdot \frac{2n+1 - (2n-1)}{(2n-1)(2n+1)}$$
$$= \frac{1}{2} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right)$$

$$n_m = a_1 + a_2 + \dots + a_m = \sum_{k=1}^m a_k = \frac{1}{2} \sum_{k=1}^m \left(\frac{1}{2k-1} - \frac{1}{2k+1} \right)$$
$$= \frac{1}{2} \left(\sum_{k=1}^m \frac{1}{2k-1} - \sum_{k=1}^m \frac{1}{2k+1} \right) = \frac{1}{2} \left(1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \dots + \frac{1}{2m-1} - \frac{1}{2m+1} \right)$$

$$= \frac{1}{2} \left(1 - \frac{1}{2m+1} \right)$$

$$s = \lim_{n \rightarrow \infty} n_m = \frac{1}{2} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2}$$

$$b) \quad a_n = \frac{1}{(n+1)(n+2)n} = \frac{1}{2} \cdot \frac{n+2 - n}{n(n+1)(n+2)} = \frac{1}{2} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right)$$

$$\begin{aligned}
 0_n &= \sum_{k=1}^n a_k = \frac{1}{2} \left(\sum_{k=1}^n \frac{1}{k(k+1)} - \sum_{k=1}^n \frac{1}{(k+1)(k+2)} \right) \\
 &= \frac{1}{2} \left(\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 3} - \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right) \\
 &= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{(n+1)(n+2)} \right)
 \end{aligned}$$

$$0 = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{1}{2} - \frac{1}{(n+1)(n+2)} \right) = \frac{1}{4} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \frac{1}{4}$$

$$\begin{aligned}
 c) \quad \frac{1}{n(n+1)\dots(n+p)} &= \frac{1}{p} \left(\frac{1}{n(n+1)\dots(n+p-1)} - \frac{1}{(n+1)\dots(n+p)} \right) \\
 &= \sum_{k=1}^n \frac{1}{k(k+1)\dots(k+p-1)} - \frac{1}{(k+1)\dots(k+p)}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{1 \cdot 2 \cdot \dots \cdot p} - \frac{1}{2 \cdot 3 \cdot \dots \cdot (p+1)} - \frac{1}{2 \cdot 3 \cdot \dots \cdot (p+1)} \\
 &\frac{1}{n} \frac{1}{1 \cdot 2 \cdot \dots \cdot p} - \frac{n!}{(n+p)!} \quad \frac{1}{p} \left(\frac{1}{p!} - \frac{n!}{(n+p)!} \right) \\
 &\frac{1}{p!} \quad \frac{1}{n \cdot p!} \left(\sum_{n=1}^{\infty} \frac{1}{n \cdot (n+p)!} \right) = \frac{1}{p \cdot p!}
 \end{aligned}$$

$$H_n: \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+3)}$$

$$\sum_{n=1}^{\infty} \frac{1}{n \cdot (n+p)!} = \frac{1}{p \cdot p!}$$

2) Let $(a_n)_{n \geq 1}$ $a_{n+1} = a_n^2 - a_n + 1, \forall n \geq 1$
 $a_1 > 1$

Prove that $\lim_{n \rightarrow \infty} a_n = \infty$ and find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{a_n}$

If $\sum_{n=1}^{\infty} a_n$ is convergent $\Rightarrow \lim_{n \rightarrow \infty} a_n = 0$

(Criterion de divergence)

If the $\lim_{n \rightarrow \infty} a_n \neq 0 \Rightarrow \sum_{k=1}^n a_k$ is divergent

ex. $\sum_{n=1}^{\infty} \frac{1}{n}$ divergent $= \infty$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Monotonicity: $a_{n+1} - a_n = a_n^2 - 2a_n + 1$
 $= (a_n - 1)^2 > 0$

$\Rightarrow a_n$ is strictly increasing $\Rightarrow a_{n+1} > a_n \Rightarrow$

$$\exists \ell = \lim_{n \rightarrow \infty} a_n \in \overline{\mathbb{R}}$$

We suppose $\ell \in \mathbb{R}$ and we test with the limit in the recurrence relation

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} (a_n^2 - a_n + 1)$$

$$l = l^2 - l + 1 \Leftrightarrow l^2 - 2l + 1 = 0 \Leftrightarrow (l-1)^2 = 0$$

$$\Rightarrow l = 1 \mid \Rightarrow l \neq 1, \text{ False} \Rightarrow \lim_{n \rightarrow \infty} a_n = +\infty$$

$a_1 > 1$
 $a_{m+1} > a_m$

$$\sum_{n=1}^{\infty} \frac{1}{a_n} = \frac{1}{a_{m+1}-1} = \frac{1}{a_m(a_m-1)}$$

$$a_{m+1} = a_m^2 - a_{m+1} = \frac{1}{a_m-1} - \frac{1}{a_m}$$

$$a_{m+1} - 1 = a_m(a_m - 1)$$

$$\Rightarrow \frac{1}{a_m} = \frac{1}{a_m-1} - \frac{1}{a_{m+1}-1}$$

$$n_m = \sum_{k=1}^m \frac{1}{a_k} = \sum_{k=1}^m \frac{1}{a_k-1} - \frac{1}{a_{k+1}-1} = \frac{1}{a_1-1} - \frac{1}{a_{m+1}-1}$$

$$n = \lim_{n \rightarrow \infty} n_m = \frac{1}{a_1-1} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{a_n} = \frac{1}{a_1-1}$$

3) Calculate the sum of the following series:

a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \rightarrow$ alternating harmonic series

b) $\sum_{n=1}^{\infty} \frac{n^3}{n!}$

a) $a_n = \frac{(-1)^{n-1}}{n} \quad n_m = ?$

(n_{2m})
 $(n_{2m+1}) \mid \rightarrow$ subsequence of n_m

$$a_{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{2n}$$

$$j_n^e = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n$$

$$a_{2n} = 1 + \frac{1}{3} + \dots + \frac{1}{2n-1} - \frac{1}{2} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right)$$

$$a_{2n} = 1 + \frac{1}{2} + \dots + \frac{1}{2n} - \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right)$$

$$a_{2n} = j_{2n}^e + \ln 2n - \ln n - j_n^e = j_{2n}^e - j_n^e + \ln 2$$

$$\lim_{n \rightarrow \infty} a_{2n} = j - j + \ln 2 = \ln 2 \quad (1)$$

$$a_{2m+1} = a_{2m} + \frac{(-1)^{2m+1-1}}{2m+1} = a_{2m} + \frac{1}{2m+1}$$

$$\lim_{n \rightarrow \infty} a_{2m+1} = \ln 2 \quad (2)$$

From (1) and (2) $\Rightarrow \lim_{n \rightarrow \infty} a_n = \ln 2$

(two subseq have the same limit and the limit is unique)

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln 2$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$

$$b) \sum_{n=1}^{\infty} \frac{n^3}{n!}, \quad a_n = \frac{n^3}{n!} = \frac{n^2}{(n-1)!}$$

$$\sum_{n=1}^{\infty} \frac{n^2-1+1}{(n-1)!} = \sum_{n=1}^{\infty} \frac{n^2-1}{(n-1)!} + \sum_{n=1}^{\infty} \frac{1}{(n-1)!}$$

For $n=1 \Rightarrow 1 + \frac{1}{1!} + \frac{1}{2!} + \dots = e$

$$\sum_{n=1}^{\infty} \frac{(n-1)(n+1)}{(n-1)!}, \text{ For } n=1 \Rightarrow 0$$

$$\sum_{n=2}^{\infty} \frac{n+1}{(n-2)!} = \sum_{n=2}^{\infty} \frac{n-2+3}{(n-2)!} = \sum_{n=3}^{\infty} \frac{1}{(n-3)!} + \sum_{n=2}^{\infty} \frac{3}{(n-2)!}$$

$$n=2 \Rightarrow 0 = e + e + 3e = (5e+3)$$

Hv: $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+3)}, a_n = \frac{1}{n(n+1)(n+3)}$

$$\frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+3} = \frac{1}{n(n+1)(n+3)} \Rightarrow A(n+1)(n+3) + nB(n+3) + Cn(n+1) = 1$$

$$+ Cn(n+1) = 1, n=-1 \Rightarrow B = -\frac{1}{2}$$

$$n=0 \Rightarrow A = \frac{1}{3}$$

$$n=-3 \Rightarrow C = \frac{1}{6}$$

$$\Rightarrow a_n = \frac{1}{3n} - \frac{1}{2(n+1)} + \frac{1}{6(n+3)} = \frac{1}{6} \left(\frac{2}{n} - \frac{3}{n+1} + \frac{1}{n+3} \right)$$

$$= \frac{1}{6} \left(2 \left(\frac{1}{n} - \frac{1}{n+1} \right) - \left(\frac{1}{n+1} - \frac{1}{n+3} \right) \right)$$

$$\sum_{k=1}^n \frac{1}{n+1} - \frac{1}{n+3} = \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \dots + \frac{1}{n-1} - \frac{1}{n+1} + \frac{1}{n} - \frac{1}{n+2}$$

$$+ \frac{1}{n+1} - \frac{1}{n+3} \Rightarrow r_n = \frac{1}{6} \left(2 \left(1 - \frac{1}{n+1} \right) - \left(\frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3} \right) \right)$$

$$r = \lim r_n = \frac{1}{6} \left(2 - \frac{5}{6} \right) = \frac{7}{36} \Rightarrow \sum_{n=1}^{\infty} r_n = \frac{7}{36}$$